

Django Apps & the Model Registry

core/utils.py::build_model_registry() scans nine domain apps; every generic endpoint is driven by the registries

Django Vector Raster WMS

DOMAIN APPS (ALLOWED_APPS) – MODELS · CALCULATIONS.PY · VIEWS

administrative (5) Province › City › District › Neighborhood PopulationProjection	physicalEnv (16) LandCover · DEM · DSM soil map · groundwater HILUCS · soil.py (SCS)	watersupply (17) extraction · treatment pipes · NRW · OPEX SupplySecurity	housing (8) supply/demand · HPI mortgage · rentals affordability	urban_heat (11) UTCI · PET · Tmrt LST · SVF · SUHII NBS · WBGT	builtup (6) ZoningArea (HILUCS) Street · Park · Facility Building · Property	nature (7) green / blue layers · trees	weather (7) WMS layers time steps	Energy (3) energy labels
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Number tags = model classes per app. Indicator dashboards (watersupply, housing, urban_heat, builtup) follow calculations.py → views._build_indicators() → HTMX partial; MOCK_DATA fills in when a province is missing.

DAG edges documented in calculations.py docstrings → core/DAG.dot (DPSIR)

scan models

build_model_registry()
core/utils.py – scans allowed_apps at startup

has GeometryField, no RasterField → VECTOR
has RasterField → RASTER · 'WMS' in key → WMS

MODEL_REGISTRY

all models

importer

field mapping, upsert keys

VECTOR_REGISTRY

GeoJSON, bounds, editing

mainMap

/api/layers/, GeoJSON, editing

RASTER_REGISTRY

COG export, tile URLs

core.signals

post_save → COG, cache bump

WMS_REGISTRY

WMS layer catalog

weather

WMS catalog, tile proxy